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Detailed Course

Motivation and Context

What is Data Science?

Data Science studies:

- Modeling real-world phenomena as numerical (quantitative) data
- Geometric and statistical properties of this data
- Properties of the spaces in which this data resides
- Data analysis and the development of tools for this analysis
- Underlying assumptions in the design of these tools

Relation to Machine Learning: Data Science studies the representation spaces in which Machine Learning operates.

> **Quick Review: Motivation**
>
> **Data Science = ML**: Study of representation spaces where ML operates.
> **Pipeline**: Process → Sensor → Raw Data → Representation (matrix in $\mathbb{R}^D$).
> Focus on **geometric AND statistical** properties.

Typical Data Pipeline

General Flow: Process → Sensor → Raw Data → Representation

Concrete Examples:

- Camera → image pixels → matrix
- Population → survey results → matrix
- Text documents → word occurrences → matrix
- Social network → relationships → matrix
- Environment → measurements → matrix

Representation Spaces

Formal Definition

We obtain data $X = \{x_1, \cdots, x_N\}$ where $x_i \in \Omega \subseteq \mathbb{R}^D$ for all $i \in [\![N]\!]$.

Statistical Perspective:

- X is a sample of size N from the underlying probability density function $f_X : \Omega \to \mathbb{R}^+$ with D variables
- We declare N i.i.d. random variables $\{X_i\}_{i \in [\![N]\!]}$ where $X_i \sim f_X$
- Each data point x_i is a realization (sample) of X_i

Algebraic Perspective:

- We form the matrix $X = \begin{bmatrix} | & & | \\ x_1 & \cdots & x_N \\ | & & | \end{bmatrix} \in \mathbb{R}^{D \times N}$
- The columns $X_{\cdot j} = x_j$ are the data vectors

Space Properties

The representation space Ω is generally a subset of $\mathbb{R}^D$:

- It is a **vector space** that can be equipped with an inner product $\langle \cdot, \cdot \rangle$
- The inner product induces a **norm** $\| \cdot \|$
- The norm induces a **distance function** $d(\cdot, \cdot)$
- Favorable case: (Ω, d) is a **metric space** where learning can be performed

Note: Ω is also the basis of a measurable space on which random variables can be defined.

> **Quick Review: Representation Spaces**
>
> $X = \{x_1, \dots, x_N\}$ with $x_i \in \Omega \subseteq \mathbb{R}^D$.
> **Two perspectives**:
>
> - **Statistical**: Sample from f_X, i.i.d. variables
> - **Algebraic**: Matrix $X \in \mathbb{R}^{D \times N}$
>
> **Structure**: Vector space → inner product → norm → distance → metric space.

Fundamental Questions

What happens when:

- D increases? (dimensionality)
- N increases? (number of samples)

Impact on:

- Data modeling
- Machine learning

Curse of Dimensionality and **Blessing of Dimensionality**

Curse of Dimensionality

1. Data Sparsity

Density Estimation Problem

Context:

- Population on hypercube $\Omega = [a, b]^D$
- Sample X from this population
- Objective: Estimate the empirical pdf via histogram

Method:

- Uniform quantization: b bins per dimension
- Desired quality: n samples per bin on average

Question: How many samples N are needed?

Answer:

- $D = 1 \rightarrow N \approx n \cdot b$
- $D = 2 \rightarrow N \approx n \cdot b^2$
- General $D \rightarrow N \approx n \cdot b^D$

Catastrophic Example: $b = 10$, $n = 10$, $D = 6 \rightarrow N \approx 10^7$ samples!

Estimation Quality with Fixed N

If we fix the number of samples at N:

Question: What estimation quality can we expect?

- $D = 1 \rightarrow n = \frac{N}{b} \rightarrow \mathbb{E}[P(x_i \in \text{bin}_j)] \approx \frac{1}{b}$
- General $D \rightarrow n = \frac{N}{b^D} \rightarrow \mathbb{E}[P(x_i \in \text{bin}_j)] \approx \frac{1}{b^D}$

Consequence: **Most bins are likely empty**

Quick Review: Data Sparsity

To estimate density with quality, need $N \approx n \cdot b^D$ samples (**exponential** growth).
If N is fixed: $n = N/b^D \rightarrow$ most bins **empty**.

> Example: $D = 6$, $b = 10$, $n = 10 \to N \approx 10^7$.

---

2. Empty Sphere and Spiky Cube

Volume of the Hypersphere

Formula for a hypersphere $B_D(r)$ of radius r centered at the origin:

$$\text{vol}(B_D(r)) = \frac{2r^D \pi^{D/2}}{D\Gamma(D/2)}$$

Recurrence Relation (for $D > 1$):

$$\text{vol}(B_D(r)) = \frac{2\pi r^2}{D} \text{vol}(B_{D-2}(r))$$

Important Property:

$$\frac{\text{area}(B_D(r))}{\text{vol}(B_D(r))} = \frac{\pi^{1/2} D\Gamma(D/2)}{\Gamma\left(\frac{D+1}{2}\right)}$$

Question: How does the volume of the sphere evolve as D increases?

Empty Sphere Phenomenon

Configuration: Hypersphere $B_D(r)$ inscribed in hypercube $C_D(r) = [-r, r]^D$.

Draw N **uniform samples** from the cube (i.e., from $\mathcal{U}(C_D(r))$).

Question: What proportion of samples falls within $B_D(r)$?

Calculation:

- $\text{vol}(B_D(r)) = \frac{2r^D \pi^{D/2}}{D\Gamma(D/2)}$
- $\text{vol}(C_D(r)) = (2r)^D$

Ratio:

$$\frac{\text{vol}(B_D(r))}{\text{vol}(C_D(r))} = \frac{\pi^{D/2}}{D \cdot 2^{D-1}\Gamma(D/2)}$$

Asymptotic Limit:

$$\lim_{D \to \infty} \frac{\text{vol}(B_D(r))}{\text{vol}(C_D(r))} = 0$$

Note: $\Gamma(n)$ behaves like $n! \sim n^n$

Consequences:

- All samples go to the part of the cube **outside the sphere** (the corners)
- The volume of the cube is **concentrated in its corners**
- All samples **escape from the center**
- The relative volume of the hypersphere tends to zero

The sphere is likely empty

Corners of the Hypercube

Configuration: Unit hypercube centered at $C_D(1) = [-\frac{1}{2}, \frac{1}{2}]^D$ with $\text{vol}(C_D(1)) = 1$.

Corner Properties:

- Number of corners: $n = 2^D$
- Each corner x_i has D coordinates: $|x_i(j)| = \frac{1}{2}$ for all j
- Distance to origin: $\|x_i\|_2 = \frac{\sqrt{D}}{2}$

Distance Table:

D	1	2	4	5	100
$\frac{\sqrt{D}}{2}$	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	1	$\frac{\sqrt{5}}{2}$	5

Observation: The distance to corners **increases with** $\sqrt{D}$ → corners move away from the center!

> **Quick Review: Empty Sphere & Spiky Cube**
>
> **Sphere/cube ratio**: $\frac{\text{vol}(B_D(r))}{\text{vol}(C_D(r))} = \frac{\pi^{D/2}}{D \cdot 2^{D-1}\Gamma(D/2)} \to 0$.
> Volume concentrated at **corners** of the cube (center-corner distance $= \sqrt{D}/2$).
> Uniform sampling → **empty** sphere.

---

3. Volume Distribution

For the Hypercube

Question: How is volume distributed in a shape of size r?

Volume ratio between hypercubes of side r and $(1-\varepsilon)r$ for $\varepsilon \in]0,1[$:

$$\frac{\text{vol}(C_D((1-\varepsilon)r))}{\text{vol}(C_D(r))} = \frac{((1-\varepsilon)r)^D}{r^D} = (1-\varepsilon)^D$$

Limit:

$$\lim_{D\to\infty} \frac{\text{vol}(C_D((1-\varepsilon)r))}{\text{vol}(C_D(r))} = 0$$

For the Hypersphere

Volume ratio between hyperspheres of radius r and $(1-\varepsilon)r$:

$$\frac{\text{vol}(B_D((1-\varepsilon)r))}{\text{vol}(B_D(r))} = \frac{2((1-\varepsilon)r)^D \pi^{D/2}}{D\Gamma(D/2)} \cdot \frac{D\Gamma(D/2)}{2r^D\pi^{D/2}} = (1-\varepsilon)^D$$

Limit:

$$\lim_{D\to\infty} \frac{\text{vol}(B_D((1-\varepsilon)r))}{\text{vol}(B_D(r))} = 0$$

Technical Note: $1-\varepsilon \leq e^{-\varepsilon}$ so $(1-\varepsilon)^D \leq e^{-D\varepsilon} \to 0$ as $D \to \infty$

Generalization to Any Volume

Principle: Decompose the volume into arbitrarily small (hyper-)cubic voxels.

Reduction: If we reduce the length of a voxel side by factor $(1-\varepsilon)$:

- Voxel volume reduced by $(1-\varepsilon)^D$
- Total volume reduced by $(1-\varepsilon)^D$

Volume Ratio:

$$\lim_{D\to\infty} \frac{\mathrm{vol}(V_D(r)) - \mathrm{vol}(V_D((1-\varepsilon)r))}{\mathrm{vol}(V_D(r))} = \lim_{D\to\infty} 1 - (1-\varepsilon)^D = 1$$

Fundamental Conclusion: **Volume is concentrated near the surface** of the shape

> **Quick Review: Volume Distribution**
>
> For **any shape**: Volume ratio $(1-\varepsilon)^D \to 0$ as $D \to \infty$.
> **Method**: Decomposition into hypercubic voxels.
> **Result**: Volume is **concentrated at the surface** (difference $\to 1$).

---

4. The Gaussian Egg

Density Along the Radius

Question: Given a Normal distribution $\mathcal{N}(\mu, \Sigma)$, what is its density along the radius r?

Context: The coordinates are D i.i.d. random variables $X_i \sim \mathcal{N}(0,1)$.

Integration along r = take the norm. R is the random variable associated with the radius:

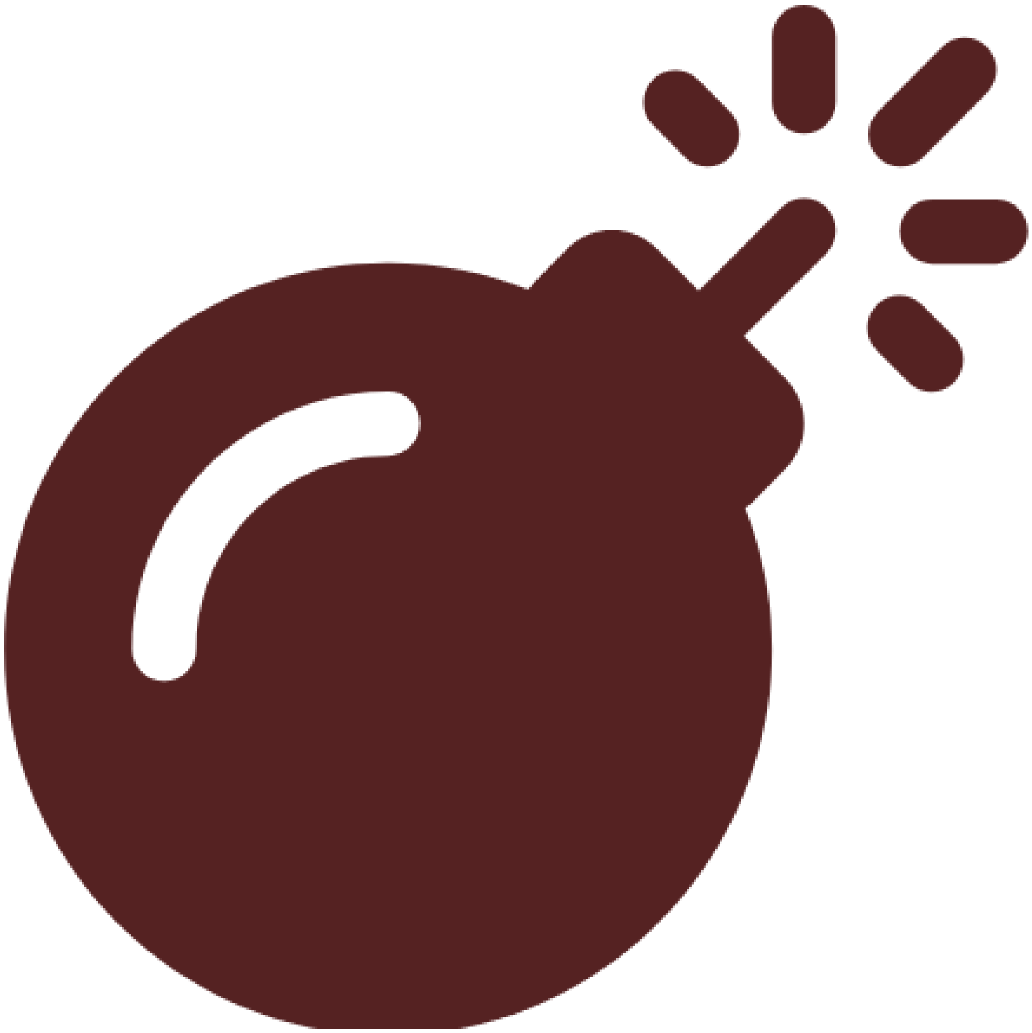
$$R^2 = \|X\|_2^2 = \sum_{i=1}^{D} X_i^2 \sim \chi^2(D)$$

Expectation:

$$\mathbb{E}[R^2] = D \implies \mathbb{E}[R] \approx \sqrt{D}$$

We can consider the centered and scaled distribution (i.e., $\mathcal{N}(0, \mathrm{Id}_D)$) with $\sigma = 1$.

Density is concentrated at radius $\sigma\sqrt{D}$



Gaussian Annulus Theorem

Given $X \sim \mathcal{N}(0, \mathrm{Id}_D)$, for any $\beta \leq \sqrt{D}$:

$$P\left[\left|\|X\|_2 - \sqrt{D}\right| > \beta\right] \leq 3e^{-c\beta^2}$$

where $c > 0$ is a constant.

Interpretation: The norm is **exponentially concentrated** around $\sqrt{D}$.

Alternative Formulation:

$$P\left[\left|\frac{\|X\|}{\sqrt{D}} - 1\right| \leq \varepsilon\right] \approx 1$$

Almost all density is in a **thin ring** around radius $\sqrt{D}$

> **Quick Review: Gaussian Egg**
>
> For $X \sim \mathcal{N}(0, \mathrm{Id}_D)$: $R^2 = \|X\|_2^2 \sim \chi^2(D)$ so $\mathbb{E}[R^2] = D$.
> **Gaussian Annulus Theorem**: $P[|\|X\|_2 - \sqrt{D}| > \beta] \leq 3e^{-c\beta^2}$.
> Density concentrated at radius $\sigma\sqrt{D}$ (**thin ring**).

---

5. Distance Concentration

Context: Data $X \subset \Omega$, select a data point x and its k nearest neighbors $V_X^k(x)$.

Let $D_{\min}$ and $D_{\max}$ be the random variables associated with the distances to the nearest and farthest neighbors among the k neighbors.

Concentration Theorem

Theorem: If $\lim_{D\to\infty} \mathrm{Var}\left[\frac{d(x,y)^p}{\mathbb{E}[d(x,y)^p]}\right] = 0$ for $0 < p < \infty$, then for any $\varepsilon > 0$:

$$\lim_{D\to\infty} P\left[\frac{D_{\max} - D_{\min}}{D_{\min}} \leq \varepsilon\right] = 1$$

Consequences:

- All neighbors appear at distance $D_{\min}$ from x
- Discriminative power **decreases exponentially**
- KNN (*k-nearest neighbors*) and εNN structures become **non-significant**

> **Quick Review: Distance Concentration**
>
> If normalized distance variance $\to 0$, then $\frac{D_{\max}-D_{\min}}{D_{\min}} \to 0$.
> All neighbors are **equidistant**.
> KNN and εNN **lose their meaning**.

---

6. Quasi-Orthogonality

Concentration of Angles Between Random Vectors

Question: When drawing vectors from a D-dimensional space, what is their expected angle?

Method: Declare 4 i.i.d. random variables W, X, Y, Z on $\Omega \subset \mathbb{R}^D$ and compute:

$$\mathbb{E}\left[\cos(\angle(X-Y, Z-W))\right] = \mathbb{E}\left[\frac{\langle X-Y, Z-W\rangle}{\|X-Y\| \cdot \|Z-W\|}\right]$$

Since W, X, Y, Z are i.i.d., $X-Y$ and $Z-W$ are also i.i.d. and centered:

$$\mathbb{E}(X-Y) = \mathbb{E}(Z-W) = 0$$

The numerator is $\operatorname{cov}(X-Y, Z-W) = 0$, the denominator is a normalizer.

Result: $\mathbb{E}[\cos(\theta)] = 0 \implies \theta \sim \frac{\pi}{2}$

Alternative:

$$\langle X-Y, Z-W\rangle = \langle X, Z\rangle - \langle X, W\rangle - \langle Y, Z\rangle + \langle Y, W\rangle$$

In expectation, all terms cancel because W, X, Y, Z are i.i.d.

Conclusion:

- Every random triangle is a **right triangle**
- Two random vectors are **quasi-orthogonal**

Concentration of Angles in a Triangle

Question: What is the expected angle in a random triangle?

Method: Declare 3 i.i.d. random variables X, Y, Z and compute for an angle:

$$\mathbb{E}\left[\cos(\angle(X-Y, Z-Y))\right] = \mathbb{E}\left[\frac{\langle X-Y, Z-Y\rangle}{\|X-Y\| \cdot \|Z-Y\|}\right]$$

Note: $X-Y$ and $Z-Y$ are **no longer independent**!

Calculation (simplified):

$$\frac{\langle X-Y, Z-Y\rangle}{\langle X-Y, X-Y\rangle} = \frac{\langle X, Z\rangle - \langle X, Y\rangle - \langle Y, Z\rangle + \langle Y, Y\rangle}{\langle X, X\rangle + \langle Y, Y\rangle - 2\langle X, Y\rangle}$$

In expectation:

$$= \frac{\langle Y, Y \rangle - \langle X, Y \rangle}{2\langle Y, Y \rangle - 2\langle X, Y \rangle} = \frac{1}{2}$$

Result: $\mathbb{E}[\cos(\theta)] = \frac{1}{2} \implies \theta \sim \frac{\pi}{3}$

Conclusion: Every random triangle is an **equilateral triangle**

Impact for the Sphere

Given the unit ball B_D, if X is a random point on its surface and Y another point, let $\theta = \angle(X, Y)$:

- If Y sampled from Ω: $\mathbb{E}[\cos(\theta)] = \frac{1}{2} \implies \theta \sim \frac{\pi}{3}$
- If Y sampled from B_D: $\mathbb{E}[\cos(\theta)] = 0 \implies \theta \sim \frac{\pi}{2}$

The volume of the sphere is contained at its **equator** (for any north direction X)

$\rightarrow$ **Alternative proof** for volume concentration at the sphere's surface

Note: To sample on the unit hypersphere, sample from a centrally symmetric distribution (e.g., Gaussian) and normalize.

> **Quick Review: Quasi-Orthogonality**
>
> **2 random vectors**: $\mathbb{E}[\cos(\theta)] = 0 \rightarrow \theta \sim \pi/2$ (quasi-orthogonal).
> **Random triangle**: $\mathbb{E}[\cos(\theta)] = 1/2 \rightarrow \theta \sim \pi/3$ (equilateral).
> **Sphere**: Volume at the equator.

---

7. Hubness

Definition

Question: How many times does a sample appear as a k-nearest neighbor of another data point?

Given X, define:

$$r_{ik}(x_j) = \begin{cases} 1 & \text{if } x_j \in V_X^k(x_i) \\ 0 & \text{otherwise} \end{cases}$$

Then:

$$n_k(x_j) = \sum_i r_{ik}(x_j)$$

Result: The distribution of n_k values is **left-skewed**.

Consequence: A small number of samples appear in the neighborhood of **many** other samples.

These **hubs** are close to all other samples

Empirical Visualization: For 20-NN with $D = 100$ and 1000 uniform samples on 50 bins, a strong asymmetry in the distribution of n_k is observed.

> **Quick Review: Hubness**
>
> $n_k(x_j)$ = number of times x_j is a k-NN of another point.
> **Asymmetric** distribution: few **hubs** appear in many neighborhoods.
> These hubs are "close" to all.

---

8. Concentration Inequalities

Fundamental Inequalities

Markov's Inequality: For a non-negative random variable X:

$$P(X > \varepsilon) \leq \frac{\mathbb{E}[X]}{\varepsilon}$$

Chebyshev's Inequality: For a random variable with mean μ and finite variance σ^2:

$$P(|X - \mu| > \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}$$

Hoeffding's Inequality

Statement: If $\{X_i\}_{i \in [\![D]\!]}$ are independent random variables with $P(a \leq X_i \leq b) = 1$ and $\mathbb{E}[X_i] = \mu$, then for any $\varepsilon > 0$:

$$P(|\bar{X} - \mu| > \varepsilon) \leq 2e^{-2D\varepsilon^2/(b-a)^2}$$

where $\bar{X} = \frac{1}{D}\sum_{i=1}^{D} X_i$

Note: Example of PAC (*Probably Approximately Correct*) analysis: probability that an event occurs approximately.

Explanation of Distance Concentration

Hoeffding's inequality explains:

- When aggregating independent variables, the aggregate **converges to their common expectation**
- The structure $\sum_{i=1}^{D}(\cdot)$ corresponds to the calculation of Minkowski distances:

$$d_p(x, y) = \left(\sum_{i=1}^{D} |x[i] - y[i]|^p\right)^{1/p}$$

- Hoeffding's inequality formally explains why distance values **concentrate**
- Although counterintuitive that all neighbors are equidistant, it is intuitive that for large D, extreme values (small or large) of $d_p(x, y)$ become unlikely: for $d_p(x, y)$ to be small (large), **all** D coordinates $x[i]$ and $y[i]$ must be (dis)similar. As D increases, this becomes unlikely.

Quick Review: Concentration Inequalities

Hoeffding: $P(|\bar{X} - \mu| > \varepsilon) \leq 2e^{-2D\varepsilon^2/(b-a)^2}$.
Aggregation of independent variables $\rightarrow$ convergence to μ.
Structure $\sum_{i=1}^{D}(\cdot)$ in distances $\rightarrow$ **concentration of distances**.

Blessing of Dimensionality

1. Laws of Large Numbers

Weak Law of Large Numbers (WLLN)

Statement: Given a set of D i.i.d. random variables $\{X_i\}_{i \in [\![D]\!]}$ with $\mathbb{E}[X_i] = \mu$, then for any $\varepsilon > 0$:

$$\lim_{D \to \infty} P(|\bar{X} - \mu| < \varepsilon) = 1$$

where $\bar{X} = \frac{1}{D} \sum_{i=1}^{D} X_i$

(Can be proven from Chebyshev's inequality)

Interpretation:

- The WLLN states that the **empirical mean** $\bar{X}$ converges (in probability) to the **true expectation** μ as the number of samples increases
- Fundamentally consistent with the frequentist approach to probabilities
- It is the **fundamental basis** of estimation theory (and density estimation)

Central Limit Theorem (CLT)

The form and rate of convergence are given by the CLT:

$$Z_D = \frac{\sqrt{D}}{\sigma}(\bar{X} - \mu) \sim \mathcal{N}(0, 1)$$

if $\mathrm{Var}(X_i) = \sigma^2$

> **Quick Review: Laws of Large Numbers**
>
> **WLLN**: $\bar{X} \xrightarrow{P} \mu$ as $D \to \infty$.
> Basis for density estimation.
> **CLT**: $Z_D = \frac{\sqrt{D}}{\sigma}(\bar{X} - \mu) \sim \mathcal{N}(0, 1)$ (convergence rate).

---

2. Random Projections

Principle

Recall: Two random vectors u and v are likely orthogonal.

Quasi-orthogonality: $\langle u, v \rangle = \varepsilon$ (small).

Key Property: **Exponentially many** quasi-orthogonal directions in high-dimensional spaces.

Application: Create random bases (quasi-orthonormal) for signal decomposition and coding.

Johnson-Lindenstrauss Lemma

Statement: Given $X \subset \Omega$, there exists a linear map $p : \mathbb{R}^D \to \mathbb{R}^d$ such that, with $0 < \varepsilon < 1$ and for all $i, j \in [\![N]\!]$:

$$(1-\varepsilon)\|x_i - x_j\| \leq \|p(x_i) - p(x_j)\| \leq (1+\varepsilon)\|x_i - x_j\|$$

and

$$d = O\left(\frac{\log N}{\varepsilon^2}\right) \leftarrow \text{Does NOT depend on } D!$$

Applications:

- *Locally Sensitive Hashing* (LSH) for approximate nearest neighbor (ANN) search
- **Caveat**: Convergence rate slower than most concentrations

> **Quick Review: Random Projections**
>
> **Principle**: Exponentially many quasi-orthogonal directions in high dimensions.
> **Johnson-Lindenstrauss**: Projection $\mathbb{R}^D \to \mathbb{R}^d$ preserving distances (up to ε) with $d = O(\log N/\varepsilon^2)$ (independent of D!).
> Application: LSH for ANN.

Importance in Data Science

1. Sampling

Consequences of Sparsity:

- An **exponential number of samples** is needed to cover a high-dimensional space
- Concentration inequalities (including WLLN and CLT) are **precision indicators** for density estimation
- In practical processes like rejection sampling, the **rejection rate** becomes exponentially close to 100% (cf. empty sphere)

2. Indexing

Exploiting Structure:

- Indexing exploits data structure to anticipate which parts of space Ω to visit at query time
- If all neighbors appear at **equal distance** from the center, the structure **disappears**

Spatial Partitioning Techniques (e.g., *kd-trees*):

- Attempt to partition space so that each data point is uniquely identified within a cell intersection (paths in the search tree)
- With distance concentration, cell size reaches the **sphere's radius** to cover the entire space

Search becomes **exhaustive** complexity $O(N)$

- *Hubness* makes some data points answers to **almost all queries**

3. Learning

Function Approximation

Machine Learning = discovering **function approximators**.

Problem: Given $\phi : X \to Y$ the true function mapping data X to labels Y, ML seeks an approximation $\phi_\theta \in \mathcal{F}$ of ϕ minimizing error (risk) on samples (train/test).

Theoretical Bound

If $\mathcal{F}$ is a class of c-uniformly Lipschitz functions on Ω:

$$\sup_{\phi_\theta \in \mathcal{F}} \|\phi_\theta - \phi\|_\infty \geq \frac{c\sqrt{D}N^{-1/D}}{2}\sqrt{\frac{2}{\pi e}}\left(1 + O\left(\frac{\log D}{D}\right)\right)$$

If we allow an error $c\varepsilon$ for $\varepsilon > 0$, the required number of training examples N is:

$$N \geq \frac{\varepsilon^{-D} D^{D/2}}{(2\pi e)^{D/2}}$$

Concrete Examples:

- $c = 1,\ \varepsilon = 0.1,\ D = 3 \rightarrow N \geq 74$
- $c = 1,\ \varepsilon = 0.1,\ D = 10 \rightarrow N \geq 687 \times 10^6$
- $c = 1,\ \varepsilon = 0.1,\ D = 15 \rightarrow N \geq 377 \times 10^{12}$
- $c = 1,\ \varepsilon = 0.01,\ D = 10 \rightarrow N \geq 6.9 \times 10^{18}$

Realistic Case in ML:

- ML data: typically hundreds of features ($D \sim O(10^2)$)
- Typical desired error: $\varepsilon \approx 10^{-4}$

$$N \geq \frac{10^{4\cdot 100} \cdot 100^{50}}{17.07^{50}} \approx 10^{400}$$

Conclusion: This is **impossible** in practice!

Quick Review: Importance in Data Science

Sampling: Exponential sample requirement, rejection rate $\rightarrow$ 100%.
Indexing: Structure disappears, kd-trees $\rightarrow$ complexity $O(N)$, hubness.
Learning: $N \geq \varepsilon^{-D} D^{D/2}/(2\pi e)^{D/2} \rightarrow$ impossible for large D and small ε.

---

4. Why Does Data Science Still Work?

Question: Why do data science processes work despite these theoretical barriers?

Assumptions in the Above Analysis:

- Most analyses assume a **uniform** (or isotropic – Normal, …) distribution of data
- Most analyses assume **independent** features

Reality:

- **Real data does NOT populate the feature space uniformly** (if features are informative)
- **Features show correlations**

In practice, data populates **subspaces** of Ω with **lower local intrinsic dimensionality**

Solutions:

- **Decorrelate features** (e.g., via linear latent space)
- **Discover data subspaces** (clustering, nonlinear dimensionality reduction)

> **Quick Review: Why It Works**
>
> Analyses assume **uniform** distribution and **independent** features.
> **Reality**: Data lies on **subspaces** of lower intrinsic dimensionality.
> **Solutions**: Decorrelation, clustering, dimensionality reduction.

Global Summary

High-dimensional geometry ($D \gtrsim 10$) is **different** from familiar 3D geometry.

+ Positive Aspects

- **Concentration inequalities** are the basis of statistical estimation
- **Laws of Large Numbers** enable density estimation
- **Random projections** exploit quasi-orthogonality

- Negative Aspects (Curse)

- Data becomes very **sparse** unless using exponentially more data
- Even with dense data, **distance and angle values concentrate**
  - From a query, all neighbors are **equidistant**
  - From a direction, everything else appears **orthogonal**
- Density concentrates on **narrow radii** → biased sampling
- **Hubs** are part of all class/query responses

Solution

Since this is by construction, the only solution is to operate in **low-dimensional spaces** in some way:

- Dimensionality reduction
- Subspace discovery
- Feature decorrelation

Quick Review: Summary

High dimension ($D \gtrsim 10$) 3D geometry.
: Concentration inequalities for estimation, random projections.
: Data sparsity, distance/angle concentration, hubs.
Solution: Operate in **low dimension**.